EE 475 Project Report

Optimization-Based Approach to Blind Image Deblurring Using Dictionary Learning

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Introduction to Image Processing | EE 475

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1 Introduction

Image deblurring is concerned with various aspects of daily life ranging from daily photography to the medical imaging. Blur can stem from many sources such as measurement equipment (e.g. de-focus blur) or the motion during the exposure time. In order to address this imaging problem, a plethora of methods have been proposed including both classical [1] and deep-learning based methods [2]. Either classical or deep-learning based methods, many of existing studies rely on harnessing the statistical properties of the image and try to address this inverse problem using Bayesian approaches. The blur which is actuated on the image is usually modeled with the convolution $y = x * h + n$ where x is the clear image, h is the blur kernel, n denotes the noise, which eventually results in the blurred-noisy output y. When both the hidden x and the kernel h is unknown, one of the most common way to find the optimal solution is to make use of MAP approach, trying to optimize the posterior distribution:

$$\begin{aligned}(x^*, h^*) &= \underset{x,h}{\operatorname{argmax}} p(x, h \mid y) \\ &= \underset{x,h}{\operatorname{argmax}} p(y \mid x, h) p(x) p(h)\end{aligned}$$

However, as Levin [3] mentions, $x = y$, $h = \delta$ solution is collapsed as a trivial one and no de-blurring occurs. To prevent this, we need to adjust the objective function so that this trivial solution is avoided.

The purpose of the project is to devise an algorithm that de-blurs the blurry image and obtain its clear version without having to know the blur kernel. The algorithm we have developed is based on initially estimating the blur kernel by learning the distribution of the kernel space and sampling from its posterior distribution, to initiate the multi-scale kernel estimation based on the proposition of Cho and Lee [4].

In order to better make use of the large dataset (whose details are mentioned in the Materials section), we learn a dictionary from the patches of clear images, which is claimed to represent the space where the clear images reside. This dictionary is then utilized to solve the constrained optimization problem that our methodology aligns with. In order to evade high dimensionality owing to the image size and large dataset, we make use of frequency space and sparse representations with online dictionary learning as proposed by Mairal in [5]. Our method incorporates various optimization techniques to build a structure that aims at resolving the de-blurring problem end-to-end.

Finally, we evaluate the performance of our model based on visual metrics depicting gradual evolution of our iterative algorithm and power distribution in the frequency domain of the clear/blurred and deblurred images

2 Materials

The proper learning of blur kernel requires clear-blurred data pairs. There are many online available datasets to choose from such as [6], [7]. The programming language to extract data and to run the algorithm is Python. Our focus is on [6] which includes clear image and clear-motion blur pairs.

Since, motion blur can vary more dramatically than the de-focus blur. Hence, motion blur can include more information and can capture individualistic differences, which are reflected on the motion blur kernels. Such a choice enabled us to use the proposition of Cho and Lee in their work [4].

3 Methodology and Algorithm

3.1 Estimating Blur Kernel

3.1.1 Initial Estimation via Sampling From Posterior Kernel Distribution

We have our dataset pairs $\{(x_m^{\text{tr}}, y_m^{\text{tr}})\}_{m=1}^{M}$ where y is blurred and x is sharp image. Patch size is $p \times p$, kernel size is denoted by $k \times k$. Vectorized kernel is $h \in \mathbb{R}^{n_h}$ with $n_h = k^2$. Our goal is to build the prior of the kernel distribution based on those extracted from the paired dataset's patches. To do so, on patches $\{(x_{m,j}, y_{m,j})\}_{j=1}^{J_m}$, $\quad x_{m,j}, y_{m,j} \in \mathbb{R}^{p \times p}$ we will try to minimize the loss function depending on the kernel as follows:

$$J(h) = \frac{1}{2}\|y - h * x\|_F^2 + \frac{\lambda}{2}\|h\|_F^2.$$

where

- y : blurred image

- x : sharp image

- h : blur kernel (to be estimated)

- (*) : convolution

- $\|\cdot\|_F$: Frobenius norm (ensuring invertibility)

Hence, for a patch we assume locally $y_{m,j} \approx h_{m,j} * x_{m,j}$ convolution model.

$$X(\omega) = \mathcal{F}\{x\}, \quad Y(\omega) = \mathcal{F}\{y\}, \quad H(\omega) = \mathcal{F}\{h\}$$

$$r = y - h * x$$

$$R(\omega) = Y(\omega) - H(\omega)X(\omega).$$

we will use the Parseval relation

$$\|r\|_F^2 = \sum_{i,j} |r[i,j]|^2 \propto \sum_{\omega} |R(\omega)|^2 = \sum_{\omega} |Y(\omega) - H(\omega)X(\omega)|^2,$$

and similarly

$$\|h\|_F^2 = \sum_{i,j} |h[i,j]|^2 \propto \sum_{\omega} |H(\omega)|^2.$$

$$J(h) = \frac{1}{2}\sum_{\omega} |Y(\omega) - H(\omega)X(\omega)|^2 + \frac{\lambda}{2}\sum_{\omega} |H(\omega)|^2.$$

frequency-wise approach:

$$J(h) = \sum_{\omega} J_{\omega}(H(\omega)),$$

where $J_{\omega}(H) = \frac{1}{2}|Y(\omega) - HX(\omega)|^2 + \frac{\lambda}{2}|H|^2$.

for the complex derivative rules, to get the ideal expression for H,

$$\frac{\partial}{\partial H^*}\left(|Y - HX|^2 + \lambda|H|^2\right) = (HX - Y)X^* + \lambda H = 0.$$

Optimal solution is then:

$$H_{m,j}(\omega) = \frac{Y_{m,j}(\omega)\overline{X_{m,j}(\omega)}}{|X_{m,j}(\omega)|^2 + \lambda}$$

We then enforce the PSF constraint

$$h_{m,j} \leftarrow \max\left(h_{m,j}, 0\right), \quad h_{m,j} \leftarrow \frac{h_{m,j}}{\sum h_{m,j}}$$

Now, we define the discrete posterior (dirac kernel buildup)

$$p(h) = \sum_{i=1}^{N} w_i^{\text{prior}}\, \delta\left(h - h_i\right), \quad \sum_i w_i^{\text{prior}} = 1$$

where weights are based on an error function

$$E_{m,j}^{\text{data}} = \frac{1}{p^2}\left\|\hat{y}_{m,j} - y_{m,j}\right\|_2^2, \quad E_{m,j}^{\text{reg}} = \lambda\left\|h_{m,j}\right\|_2^2,$$

$$\ell_{m,j} = E_{m,j}^{\text{data}} + E_{m,j}^{\text{reg}}.$$

based on which we build Boltzmann weights as a natural choice

$$w_i^{\text{prior}} = \frac{\exp\left(-\beta_{\text{tr}}\tilde{\ell}_i\right)}{\sum_{j=1}^{N}\exp\left(-\beta_{\text{tr}}\tilde{\ell}_j\right)}$$

Then, we apply posterior update per test patch using feature similarity. From test blurred image y^*, we extract non-overlapping patches $\{y^*(k)\}_{k=1}^{K}$ for which we define a feature vector $\phi_k^* = \phi\left(y^*(k)\right)$ where

$$\phi = \begin{bmatrix} R_{HL}(y) \\ b(y) \\ \mu_g(y) \\ \text{Var}(|\nabla y|) \end{bmatrix} \in \mathbb{R}^4$$

where high/low radial power ratio R_{HL} is defined $R_{HL}(y) = \frac{\sum_{r(\omega)>r_0} P(\omega)}{\sum_{r(\omega)\le r_0} P(\omega)+\varepsilon}$ with $r_0 = 0.25\max r(\omega)$ and $P(\omega) = |\mathcal{F}\{y\}(\omega)|^2$.

The other feature is slope of log-power vs radius $\log P(r) \approx a + br$ and we store b value.

The other features are related to the gradient information's first two order statistics: $\mu_g(y) = \mathbb{E}[|\nabla y|]$ and $\mathrm{Var}(|\nabla y|) = \mathbb{V}[|\nabla y|]$.

We use a gaussian-like similarity based likelihood:

$$p\left(\phi_k^* \mid h_i\right) \propto \exp\left(-\gamma \left\|\phi_i - \phi_k^*\right\|_2^2\right)$$

Then the posterior is

$$p\left(h = h_i \mid \phi_k^*\right) \propto p\left(\phi_k^* \mid h_i\right) p\left(h = h_i\right)$$

$$w_{i,k}^{\text{post}} = \frac{w_i^{\text{prior}} \exp\left(-\gamma d_{i,k}^2\right)}{\sum_{j=1}^{N} w_j^{\text{prior}} \exp\left(-\gamma d_{j,k}^2\right)}$$

$$p\left(h \mid \phi_k^*\right) = \sum_{i=1}^{N} w_{i,k}^{\text{post}} \, \delta\left(h - h_i\right)$$

Then we apply sampling from kernel density (KDE) based regularized posterior distribution to get posterior samples for each patch. Since we assume that the image is blurred by a single kernel, we average these candidates. The following algorithms depict this initialization step:

Algorithm 1 Stage 1A: Kernel Extraction, Reconstruction Loss, and Blur-Feature Computation

Input: Training pairs $\{(x_m^{\mathrm{tr}}, y_m^{\mathrm{tr}})\}_{m=1}^{M}$;
Patch size p, kernel size k; ridge parameter λ.
Output: Kernel set $\mathcal{H}$, loss set $\mathcal{L}$, feature set Φ.

Initialize $\mathcal{H} \leftarrow \emptyset$, $\mathcal{L} \leftarrow \emptyset$, $\Phi \leftarrow \emptyset$
for $m = 1, \ldots, M$ **do**
  Extract overlapping patches $\{(x_{m,j}, y_{m,j})\}_{j=1}^{J_m}$
  for $j = 1, \ldots, J_m$ **do**
    `// 1.1 Kernel estimation via frequency inversion`
    $X_{m,j} \leftarrow \mathcal{F}\{x_{m,j}\}, \quad Y_{m,j} \leftarrow \mathcal{F}\{y_{m,j}\}$

$$H_{m,j}(\omega) = \frac{Y_{m,j}(\omega)\,\overline{X_{m,j}(\omega)}}{|X_{m,j}(\omega)|^2 + \lambda}$$

    Inverse FFT, center, crop $\Rightarrow h_{m,j} \in \mathbb{R}^{k \times k}$ $h_{m,j} \leftarrow \max(h_{m,j}, 0), \quad h_{m,j} \leftarrow h_{m,j} / \sum h_{m,j}$
    `// 1.2 Patch reconstruction loss`
    $\hat{y}_{m,j} = h_{m,j} * x_{m,j}$

$$E_{m,j}^{\mathrm{data}} = \frac{1}{p^2}\|\hat{y}_{m,j} - y_{m,j}\|_2^2, \qquad E_{m,j}^{\mathrm{reg}} = \lambda \|h_{m,j}\|_2^2,$$

$$\ell_{m,j} = E_{m,j}^{\mathrm{data}} + E_{m,j}^{\mathrm{reg}}.$$

    `// 1.3 Four-dimensional blur feature vector`
    Power spectrum $P = |\mathcal{F}\{y_{m,j}\}|^2$, radial frequencies $r(\omega)$ $r_0 = 0.25 \max r(\omega)$

$$R_{\mathrm{HL}}(y_{m,j}) = \frac{\sum_{r(\omega) > r_0} P(\omega)}{\sum_{r(\omega) \le r_0} P(\omega) + \varepsilon}$$

    Fit slope $b(y_{m,j})$ from $\log P(r) \approx a + b\, r$
    Compute gradient magnitude statistics:

$$\mu_g(y_{m,j}) = \mathbb{E}[|\nabla y_{m,j}|], \qquad \mathrm{Var}(|\nabla y_{m,j}|) = \mathbb{V}[|\nabla y_{m,j}|].$$

$$\phi_{m,j} = \begin{bmatrix} R_{\mathrm{HL}}(y_{m,j}) \\ b(y_{m,j}) \\ \mu_g(y_{m,j}) \\ \mathrm{Var}(|\nabla y_{m,j}|) \end{bmatrix} \in \mathbb{R}^4.$$

    Append $h_{m,j}$, $\ell_{m,j}$, $\phi_{m,j}$ to $\mathcal{H}, \mathcal{L}, \Phi$
Let $N = |\mathcal{H}|$ and index by h_i, ℓ_i, ϕ_i

Algorithm 2 Stage 1B: Prior Weight Construction

Input: Losses $\{\ell_i\}_{i=1}^{N}$; prior scale β_{tr}.
Output: Prior weights $\{w_i^{\mathrm{prior}}\}_{i=1}^{N}$.

Shift losses:

$$\tilde{\ell}_i = \ell_i - \min_j \ell_j.$$

Compute prior weights:

$$w_i^{\mathrm{prior}} = \frac{\exp(-\beta_{\mathrm{tr}} \tilde{\ell}_i)}{\sum_j \exp(-\beta_{\mathrm{tr}} \tilde{\ell}_j)}.$$

Algorithm 3 Stage 1C: KDE Bandwidth Computation

Input: Number of kernels N; kernel dimension $n_h = k^2$; bandwidth constant A.
Output: KDE bandwidth h_{opt}.

Compute:

$$h_{\text{opt}} = A\, N^{-1/(n_h+4)}.$$

Algorithm 4 Stage 2: Posterior Update and Jittered Sampling

Input: Test image y^*; patch size p; kernel size k; feature scale γ;
Kernel set $\mathcal{H}$, feature set Φ, prior weights w_i^{prior},
KDE bandwidth h_{opt}.
Output: Patch-level kernels $\{\tilde{h}_k\}_{k=1}^K$.

Extract non-overlapping test patches $\{y^*_{(k)}\}_{k=1}^K$
for $k = 1, \ldots, K$ **do**
  Compute test feature vector ϕ_k^* (same 4 components)
  `// Posterior weights`

$$d_{i,k}^2 = \|\phi_i - \phi_k^*\|_2^2, \qquad w_{i,k}^{\text{post}} \propto w_i^{\text{prior}} e^{-\gamma d_{i,k}^2}.$$

  Normalize
  `// Weighted mean + covariance`

$$\mu_k = \sum_i w_{i,k}^{\text{post}} h_i, \qquad S_k = \sum_i w_{i,k}^{\text{post}} (h_i - \mu_k)(h_i - \mu_k)^\top + \varepsilon I.$$

  Compute $S_k = D_k D_k^\top$
  `// Discrete ancestor + Gaussian jitter`
  Sample $a_k \sim \text{Categorical}(w_{\cdot,k}^{\text{post}})$ Draw $\varepsilon_k \sim \mathcal{N}(0, I_{n_h})$

$$\tilde{h}_k = h_{a_k} + h_{\text{opt}} D_k \varepsilon_k.$$

  Reshape to $k \times k$, enforce nonnegativity and sum-to-one
  Store $\tilde{h}_k$

Algorithm 5 Stage 3: Aggregate Patch Kernels

Input: Patch kernels $\{\tilde{h}_k\}_{k=1}^K$.
Output: Final kernel $\hat{h}$.

$$\hat{h} = \frac{1}{K} \sum_{k=1}^K \tilde{h}_k, \qquad \hat{h} \leftarrow \max(\hat{h}, 0), \quad \hat{h} \leftarrow \hat{h} / \sum \hat{h}.$$

Output: $\hat{h}$

3.1.2 Fine Tune with Multi-Scale Kernel Estimation

Now that we have an initial estimate we make use of the proposition of Cho and Lee [4]. They solve this $y = h * x + N$ equation in alternating steps:

$$X' = \text{argmin}_X \left\{ \|Y - H * X\| + \rho_X(X) \right\},$$
$$H' = \text{argmin}_H \left\{ \|Y - H * X\| + \rho_H(H) \right\}.$$

to which they integrate coarse-to-fine scheme. At the coarsest level, we use the down-sampled version of h to initialize the process with the prediction step. After the final estimate of x has been obtained at a coarse level, it is up-sampled by bilinear interpolation and then used for the input of the first prediction step at the next finer level.

Their prediction step consists of bilateral filtering, shock filtering, and gradient magnitude thresholding. They first apply bilateral filtering to the current estimate of X to suppress possible noise and small details. A shock filter is then used to restore strong edges of X. The result X' of shock filtering contains not only high-contrast edges but also enhanced noise. We remove the noise by computing and thresholding the gradient maps $\{\partial_x X', \partial_y X'\}$ of X'. The truncated gradient maps $\{P_x, P_y\}$ give the final output of the prediction step.

Here are the mathematical steps:

We build a pyramid

$$B^{(s)} = \text{Resize}\left(B_{\text{full}}, s \times s\right), \quad s \in \{32, 64, 128, 256\}$$

at the coarsest scale $s = 32$ we initialize $L^{(32)} \leftarrow B^{(32)}$ and the kernel $k^{(5)}$ (by posterior sample)

Then for each scale we : - Upsample L to the new size

- Resize kernel to the new kernel size ($5 \to 9 \to 17 \to 33$)

- Run N iterations (e.g. 12) of Prediction $\to$ Kernel update $\to$ Latent update

Prediction

This is Cho-Lee's "prediction step": bilateral smoothing, shock filtering and keep-only-strong-edges step. Bilateral filtering $X_b = \text{Bilateral}\left(X; \sigma_s, \sigma_r\right)$ reduces noise but preserves edges. Shock filter sharpens the edges $X \leftarrow X - \text{sign}(\Delta X)|\nabla X|\Delta t$ where ΔX is the Laplacian and $|\nabla X|$ is gradient magnitude. For the strong edge mask we calculate

$$g_x = \partial_x X_s, \quad g_y = \partial_y X_s, \quad g = \sqrt{g_x^2 + g_y^2}$$

threshold τ so that only a fraction of pixels survive:

$$\mathcal{M} = \{(i, j) : g(i, j) \geq \tau\}$$

$$P_x = g_x \cdot \mathbf{1}_{\mathcal{M}}, \quad P_y = g_y \cdot \mathbf{1}_{\mathcal{M}}$$

where P_x, P_y (sparse derivative maps)

Kernel Update

We build the blurred derivatives $\partial_x Y, \partial_y Y, \partial_{xx} Y, \partial_{yy} Y, \partial_{xy} Y$ and first and second order derivative pairs $(P_x, \partial_x Y)$ with weight w_1, $(P_y, \partial_y Y)$ with weight w_1 ,$(\partial_x P_x, \partial_{xx} Y)$ with weight w_2, $(\partial_y P_y, \partial_{yy} Y)$ weight w_2, $(P_{xy}, \partial_{xy} Y)$ with weight w_2 where $P_{xy} = \frac{1}{2}\left(\partial_x P_y + \partial_y P_x\right)$.

The objective function is adopted as

$$\min_k \sum_m w_m \|P_m * h - Y_m\|_2^2 + \beta \|h\|_2^2$$

where m ranges over the 5 derivative channels, which yields a closed form solution in the frequency domain.

$$\hat{H}(\omega) = \frac{\sum_m w_m \overline{\hat{P}_m} \hat{Y}_m}{\sum_m w_m \left|\hat{P}_m\right|^2 + \beta}$$

For the latent clear image estimation the objective function is

$$\min_X \underbrace{w_0 \|h * X - Y\|_2^2}_{\text{DC anchor}} + \underbrace{w_1 \sum_{d\in\{x,y\}} \|h * \partial_d X - \partial_d Y\|_2^2}_{\text{1 st order}} + \underbrace{w_2 \sum_{d\in\{xx,yy,xy\}} \|h * \partial_d X - \partial_d Y\|_2^2}_{\text{2 nd order}} + \alpha \|\nabla X\|_2^2$$

The w values are adopted from the values presented in [13] as Cho and Lee did in their work. We can derive its closed-form update solution in frequency domain as follows:

Knowing that the derivative and the convolution has the following commutation relation: $h * (\partial X) = (\partial(h * X)) = (\partial h) * X$. Let $\widehat{D_x}(\omega)$ be the FFT of the derivative filter ∂_x (and the same notation for $D_y, D_{xx}, \ldots$ which are easily derived by the finite difference in the discrete domain and hats denote the FT)

$$\widehat{h * \partial_x X}(\omega) = \widehat{h}(\omega)\widehat{\partial_x X}(\omega) = \widehat{H}(\omega)\widehat{D_x}(\omega)\widehat{X}(\omega)$$

and eventually $\widehat{h * \partial_x X - \partial_x Y}(\omega) = \widehat{D_x}(\omega)(\widehat{H}(\omega)\widehat{X}(\omega) - \widehat{Y}(\omega))$

and $\widehat{h * X - Y}(\omega) = \widehat{H}(\omega)\widehat{X}(\omega) - \widehat{Y}(\omega)$. We use the Parseval relation: $\|Z\|_2^2 \propto \sum_\omega |\widehat{Z}(\omega)|^2$

Hence the objective function (or as known as cost function) becomes

$$J_\omega(X) = W|HX - Y|^2 + \alpha M_1 |X|^2$$

where

$$W(\omega) = w_0 + w_1 M_1(\omega) + w_2 M_2(\omega)$$
$$M_1(\omega) = |D_x|^2 + |D_y|^2, \quad M_2(\omega) = |D_{xx}|^2 + |D_{yy}|^2 + |D_{xy}|^2$$

Optimization with respect to the clear image X follows a straightforward procedure of element-wise (for each frequency w as done previously) optimization:

$$J_\omega(X) = W(HX - Y)(\bar{H}\bar{X} - \bar{Y}) + \alpha M_1 X \bar{X}$$

Using the properties $\frac{\partial}{\partial \bar{X}}(X\bar{X}) = X$ and $\frac{\partial}{\partial \bar{X}}(\bar{H}\bar{X}) = \bar{H}$

$$\frac{\partial J_\omega}{\partial \bar{X}} = W(HX - Y) \cdot \bar{H} + \alpha M_1 X$$

$$W\bar{H}(HX - Y) + \alpha M_1 X = 0$$

$$\left(W|H|^2 + \alpha M_1\right) X = W\bar{H}Y$$

$$X = \frac{W\bar{H}Y}{W|H|^2 + \alpha M_1}$$

Thus,

$$\widehat{X}(\omega) = \frac{\overline{\widehat{H}(\omega)}\widehat{Y}(\omega)W(\omega)}{|\widehat{H}(\omega)|^2 W(\omega) + \alpha M_1(\omega)}$$

3.2 Dictionary Learning

Dictionary learning plays a fundamental role in our pipeline. Based on a large dataset and their patches, we can learn a representation by dictionary whose elements are rich enough to construct any clear patch in ideal scenario. Our large dataset and its patches provide us with a sufficient amount so that the elements of the dictionary can be combined (by so-called "atoms").

Dictionary will come into play in our objective function as a constraint when learning the clear latent image and this constraint will enforce the reconstruction to comply with the space of the clear images. Here are the details of how to learn such a representation.

Now we switch to vectorized form and for images of size $H_{\text{img}} \times W_{\text{img}}$, let $M = H_{\text{img}} W_{\text{img}}$.

- Clean image variable

$$z_x \in \mathbb{R}^M \text{ (vectorized clean image)}$$

- Observed blurred image

$$z_y \in \mathbb{R}^M \text{ (vectorized blurry image; data)}$$

We represent the clean image as $z_x = D_z \alpha$ where D corresponds the dictionary. We determine it before starting our main algorithm. They can be hand-crafted linear transforms (such as discrete cosine transform (DCT), Wavelet Transform Coefficients) or learned with the optimization:

$$\min_{D,\{\alpha_i\}} \sum \|x_i - D\alpha_i\|_2^2 + \lambda \|\alpha_i\|_1 \quad \text{s.t.} \quad \|d_j\|_2 = 1$$

However, the issue is that when our sample size increases, the Z matrix gets large (used through the algorithm). In order to deal with the high dimensionality, we propose to use the patch version, whose formalism is as follows:

We know that $z_x \in \mathbb{R}^M$ and we extract the patch as $p_\ell = P_\ell z_x \in \mathbb{R}^{n_p}$ where $n_p = p^2$ for $p * p$ patch. We will try to learn the dictionary on the patches as: $p_\ell \approx D\alpha_\ell$

In order to fuse these patches back, we apply the following optimization, aiming at the estimated fusion minimizes L_2 error:

$$\min_z F(z) := \sum_{\ell=1}^{L} \|P_\ell z - r_\ell\|_2^2, \quad r_\ell := D\alpha_\ell \text{ fixed}$$

$$F(z) = \sum_\ell \left(z^\top P_\ell^\top P_\ell z - 2z^\top P_\ell^\top r_\ell + r_\ell^\top r_\ell\right)$$

$$F(z) = z^\top \left(\sum_\ell P_\ell^\top P_\ell\right) z - 2z^\top \left(\sum_\ell P_\ell^\top r_\ell\right) + \text{ constant with respect to z}$$

Re-expressing:

$$F(z) = z^\top A z - 2z^\top b + \text{cons}$$

where,

$$A := \sum_\ell P_\ell^\top P_\ell, \quad b := \sum_\ell P_\ell^\top r$$

$$\nabla_z F(z) = 0 \quad \Longrightarrow \quad 2Az - 2b = 0$$

$$Az = b \quad \Longrightarrow \quad z = A^{-1}b$$

Hence, the reconstruction is based on:

$$z^* = \left(\sum_\ell P_\ell^\top P_\ell\right)^{-1} \left(\sum_\ell P_\ell^\top r_\ell\right) = \left(\sum_\ell P_\ell^\top P_\ell\right)^{-1} \left(\sum_\ell P_\ell^\top D\alpha_\ell\right).$$

where $W = \sum_\ell P_\ell^\top P_\ell$ and hence $z = W^{-1} \sum_\ell P_\ell^\top D\alpha\ell$

The properties of W is as follows:

P_ℓ is such a matrix that it has only a single 1 entry at each row:

$$\left(P_\ell^\top P_\ell\right)_{ij} = \sum_{m=1}^{n_p} \left(P_\ell\right)_{mi} \left(P_\ell\right)_{mj}.$$

Hence, for $i \neq j$: every term in the sum is 0.

$$\left(P_\ell^\top P_\ell\right)_{ii} = \begin{cases} 1 & \text{if pixel } i \text{ belongs to patch } \ell \\ 0 & \text{otherwise .} \end{cases}$$

Knowing that

$$W := \sum_{\ell=1}^{L} P_\ell^\top P_{\ell.}$$
$$W_{ij} = \sum_{\ell} \left(P_\ell^\top P_\ell\right)_{ij.}$$

and due to diagonality:

$$W_{ii} = \sum_{\ell} \left(P_\ell^\top P_\ell\right)_i$$

Therefore W_{ii} counts how many patches include pixel i.

Conceptually:

$$\alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_L \end{bmatrix} \in \mathbb{R}^{KL}$$

$$D_{\mathrm{z}}\ \alpha = W^{-1} \sum_{\ell=1}^{L} P_\ell^\top\ (D\alpha_\ell)$$

$$D_{\mathrm{z}}\ = W^{-1} \begin{bmatrix} P_1^\top D & P_2^\top D & \cdots & P_L^\top D \end{bmatrix} \in \mathbb{R}^{M \times (KL)}$$

but as we will see, we won't need to store the large D_{z} matrix.

These W and P may look large matrices but in code implementation, we don't need to store the explicit matrices, index slicing will be harnessed.

To deal with large number of extracted patches, we can use online dictionary learning proposed in [5]. They gradually learn the dictionary as new data arrives based on the following algorithm, adopted directly from the reference paper:

Algorithm 6 Online dictionary learning.

Require: $\mathbf{x} \in \mathbb{R}^m \sim p(\mathbf{x})$ (random variable and an algorithm to draw i.i.d samples of p), $\lambda \in \mathbb{R}$ (regularization parameter), $\mathbf{D}_0 \in \mathbb{R}^{m \times k}$ (initial dictionary), T (number of iterations).

1: $\mathbf{A}_0 \in \mathbb{R}^{k \times k} \leftarrow 0$, $\mathbf{B}_0 \in \mathbb{R}^{m \times k} \leftarrow 0$ (reset the "past" information). **for** $t = 1$ *to* T **do**

2:

Draw $\mathbf{x}_t$ from $p(\mathbf{x})$.

3: **Sparse coding:** compute using Coordinate Descent

$$\boldsymbol{\alpha}_t \triangleq \arg\min_{\boldsymbol{\alpha} \in \mathbb{R}^k} \frac{1}{2}\|\mathbf{x}_t - \mathbf{D}_{t-1}\boldsymbol{\alpha}\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_1.$$

4: $\mathbf{A}_t \leftarrow \mathbf{A}_{t-1} + \boldsymbol{\alpha}_t \boldsymbol{\alpha}_t^T$.

5: $\mathbf{B}_t \leftarrow \mathbf{B}_{t-1} + \mathbf{x}_t \boldsymbol{\alpha}_t^T$.

6: Compute $\mathbf{D}_t$ using Algorithm 2, with $\mathbf{D}_{t-1}$ as warm restart, so that

$$\begin{aligned}\mathbf{D}_t &\triangleq \arg\min_{\mathbf{D} \in \mathcal{C}} \frac{1}{t}\sum_{i=1}^{t}\left(\frac{1}{2}\|\mathbf{x}_i - \mathbf{D}\boldsymbol{\alpha}_i\|_2^2 + \lambda\|\boldsymbol{\alpha}_i\|_1\right) \\ &= \arg\min_{\mathbf{D} \in \mathcal{C}} \frac{1}{t}\left(\frac{1}{2}\mathrm{Tr}(\mathbf{D}^T\mathbf{D}\mathbf{A}_t) - \mathrm{Tr}(\mathbf{D}^T\mathbf{B}_t)\right). \end{aligned} \tag{6}$$

7:

8: **return** $\mathbf{D}_T$ (learned dictionary).

Algorithm 7 Dictionary Update.

Require: $\mathbf{D} = [\mathbf{d}_1, \ldots, \mathbf{d}_k] \in \mathbb{R}^{m \times k}$ (input dictionary), $\mathbf{A} = [\mathbf{a}_1, \ldots, \mathbf{a}_k] \in \mathbb{R}^{k \times k}$, $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_k] \in \mathbb{R}^{m \times k}$ **repeat**

| $j = 1$ to k

until for do

1: Update the j-th column to optimize for (6):

$$\begin{aligned}\mathbf{u}_j &\leftarrow \frac{1}{\mathbf{A}[j,j]}(\mathbf{b}_j - \mathbf{D}\mathbf{a}_j) + \mathbf{d}_j, \\ \mathbf{d}_j &\leftarrow \frac{1}{\max(\|\mathbf{u}_j\|_2, 1)}\mathbf{u}_j. \end{aligned} \tag{7}$$

2:

3: convergence

4: **return D** (updated dictionary).

3.3 Inference

We can consider the preparation step as the training stage since we learn the optimal D and H based on the data pairs we have. Having learned these components, now we analyze the inference (i.e. what should be done when a new blurry image arrives at our model)

We assume that the vectorized noisy version is based on the $z_y \approx Hz_x$ transformation where

H is conceptually $M \times M$

$$Hz_x = \mathrm{vec}(h * X), \quad X = \mathrm{reshape}\,(z_x) \in \mathbb{R}^{H_{\mathrm{img}} \times W_{\mathrm{imx}}}$$

The optimization will include z_x, α

3.3.1 Fixed H and D

Our objective function is

$$\min_{z_x,\alpha} \frac{1}{2} \|z_y - Hz_x\|_2^2 + \lambda\|\alpha\|_1 \quad \text{s.t. } z_x = D_z\alpha$$

notation:

$$\min_{z_x,\alpha} f(z_x) + g(\alpha) \quad \text{s.t. } z_x = D_z\alpha$$

- $f(z_x) = \frac{1}{2}\|z_y - Hz_x\|_2^2$
- $g(\alpha) = \lambda\|\alpha\|_1$.
- $z_x = D_z\alpha$ is the dictionary on the entire image vector.

We will use the alternating direction method of multipliers (ADMM) as proposed in [1], [8], [9]:

$$\mathcal{L}(z_x, \alpha, \lambda) = f(z_x) + g(\alpha) + \lambda^\top (z_x - D_z\alpha) + \frac{\mu}{2}\|z_x - D_z\alpha\|_2^2.$$

$$u := \frac{\lambda}{\mu} \quad \Longleftrightarrow \quad \lambda = \mu u.$$

$$\mathcal{L}(z_x, \alpha, \lambda) = f(z_x) + g(\alpha) + (\mu u)^\top (z_x - D_z\alpha) + \frac{\mu}{2}\|z_x - D_z\alpha\|_2^2$$

Knowing that:

$$\|z_x - D_z\alpha + u\|_2^2 = \|z_x - D_z\alpha\|_2^2 + 2u^\top (z_x - D_z\alpha) + \|u\|_2^2.$$

Multiply by $\frac{\mu}{2}$:

$$\frac{\mu}{2}\|z_x - D_z\alpha + u\|_2^2 = \frac{\mu}{2}\|z_x - D_z\alpha\|_2^2 + \mu u^\top (z_x - D_z\alpha) + \frac{\mu}{2}\|u\|_2^2$$

$$\mu u^\top (z_x - D_z\alpha) + \frac{\mu}{2}\|z_x - D_z\alpha\|_2^2 = \frac{\mu}{2}\|z_x - D_z\alpha + u\|_2^2 - \frac{\mu}{2}\|u\|_2^2.$$

Then we can express it:

$$\mathcal{L}_\mu(z_x, \alpha, u) = f(z_x) + g(\alpha) + \frac{\mu}{2}\|z_x - D_z\alpha + u\|_2^2 - \frac{\mu}{2}\|u\|_2^2.$$

which is called scaled ADMM form [9]. The alternating solution is as:

$$\begin{aligned}
z_x^{k+1} &= \arg\min_{z_x} f(z_x) + \frac{\mu}{2}\left\|z_x - D_z\alpha^k + u^k\right\|_2^2 \\
\alpha^{k+1} &= \arg\min_{\alpha} g(\alpha) + \frac{\mu}{2}\left\|z_x^{k+1} - D_z\alpha + u^k\right\|_2^2 \\
u^{k+1} &= u^k + \left(z_x^{k+1} - D_z\alpha^{k+1}\right)
\end{aligned}$$

Optimizing z_x

$$\begin{aligned}
&\nabla_{z_x} f(z) = H^\dagger \left(H z_x - z_y\right) \\
&H^\dagger \left(H z_x^{k+1} - z_y\right) + \mu \left(z_x^{k+1} - D_z \alpha^k + u^k\right) = 0 \\
&\quad \Longrightarrow \left(H^\dagger H + \mu I\right) z_x^{k+1} = H^\dagger z_y + \mu \left(D_z \alpha^k - u^k\right)
\end{aligned}$$

Here, H matrix denoting the circular convolution in the vectorized form is a huge matrix with the following form [10]

$$H = \begin{bmatrix} C_0 & C_{W-1} & \cdots & C_1 \\ C_1 & C_0 & \cdots & C_2 \\ \vdots & \vdots & \ddots & \vdots \\ C_{W-1} & C_{W-2} & \cdots & C_0 \end{bmatrix}$$

where

$$(C_v)_{i,k} = h[(i-k) \bmod H, v], \quad i,k = 0, \ldots, H-1.$$

are the circulant matrix blocks corresponding 1-D convolution.

This matrix is not memory-friendly. We will use its properties to simplify the computation:

We will make use of the Fourier-related diagonalization [10] of the circulant matrices:

$$H = F^{-1} \Lambda F$$

where:

- $F = F_W \otimes F_H$ kronecker product which is the separation of 2-D DFT kernel into its axis-components and $F^{-1} = F^\dagger$

- Λ is a diagonal matrix with the DFT-2 of the kernel h :

$$\Lambda = \operatorname{diag}\left(H\left(\omega_x, \omega_y\right)\right)$$

$$H = \left(F_W \otimes F_H\right)^{-1} \operatorname{diag}\left(H\left(\omega_x, \omega_y\right)\right) \left(F_W \otimes F_H\right)$$

Hence, we will derive the update relation using these properties:

$$H^\dagger H = \left(F^\dagger \Lambda^\dagger F\right)\left(F^\dagger \Lambda F\right) = F^\dagger \Lambda^\dagger \left(F F^\dagger\right) \Lambda F = F^\dagger \left(\Lambda^\dagger \Lambda\right) F$$

Hence,

$$H^\dagger H = F^\dagger \Gamma F, \quad \Gamma := \operatorname{diag}\left(\left|H\left(\omega_k\right)\right|^2\right)$$

Knowing that we had this expression:

$$\left(H^{\dagger}H + \mu I\right) z_x = H^{\dagger} z_y + \mu b$$

where $b := D_{\mathrm{z}}\, \alpha - u$

$$\begin{aligned} b &:= D_{\mathrm{z}}\, \alpha - u \\ &= W^{-1} \sum_{\ell=1}^{L} P_{\ell}^{\top} \left(D\alpha_{\ell}\right) - W^{-1} \sum_{\ell=1}^{L} P_{\ell}^{\top} u_{\ell} \\ &= W^{-1} \sum_{\ell=1}^{L} P_{\ell}^{\top} \left(D\alpha_{\ell} - u_{\ell}\right). \end{aligned}$$

where we define:

$$u^k = W^{-1} \sum_{\ell} P_{\ell}^{\top} u_{\ell}^k$$

(Hence we don't have the find D_{z})

Equally

$$F\left(H^{\dagger}H + \mu I\right) z_x = F\left(H^{\dagger} z_y + \mu b\right).$$

$$F\left(H^{\dagger}H\right) z_x + \mu F z_x = F H^{\dagger} z_y + \mu F b.$$

Knowing that

$$F\left(H^{\dagger}H\right) z_x = F\left(F^{\dagger}\Gamma F\right) z_x = \left(F F^{\dagger}\right) \Gamma(F z_x) = \Gamma(F z_x)$$

Defining,

$$X := F z_x, \quad Y := F z_y, \quad B := F b.$$

The LHS of the equation becomes:

$$\Gamma X + \mu X = (\Gamma + \mu I) X$$

For the RHS, we re-express $F H^{\top} y + \mu F b$ as follows:

$$F H^{\dagger} y = F\left(F^{\dagger}\Lambda^{\dagger} F\right) y = \left(F F^{\dagger}\right) \Lambda^{\dagger}(F y) = \Lambda^{\dagger} Y.$$

Then the equation becomes

$$(\Gamma + \mu I) X = \Lambda^{\dagger} Y + \mu B$$

Due to Kronecker product:

$$X := Fz_x = (F_W \otimes F_H)\, x = \mathrm{vec}\left(\mathrm{FFT2}\left(X_{\mathrm{img}}\right)\right)$$

making this equation element-wise for all ω

$$\left(|H(\omega)|^2 + \mu\right) X(\omega) = \overline{H(\omega)} Y(\omega) + \mu B(\omega).$$

Then,

$$X^{k+1}(\omega) = \frac{\overline{H(\omega)} Y(\omega) + \mu B^k(\omega)}{|H(\omega)|^2 + \mu}$$

and thus

$$z_x^{k+1} = \mathrm{vec}\left(\mathcal{F}^{-1}\left\{X^{k+1}(\omega)\right\}\right)$$

Optimizing α

We cannot ensure square orthonormal dictionary ($D^\top D = I$ or $DD^\top = I$) property for online dictionary learning, so we do not have closed form solutions [5].

$\alpha^{k+1} = \arg\min_\alpha \lambda\|\alpha\|_1 + \frac{\mu}{2}\left\|z_x^{k+1} - D_z\alpha + u^k\right\|_2^2$ is solved by ISTA [11], [12] as follows:

1. Gradient step

Define: $m^k = z_x^{k+1} + u^k$

$$\tilde{\alpha}_{t+1}^k = \alpha_t^k - \frac{1}{L}\mu D^\top\left(D\alpha_t^k - m^k\right).$$

2. Soft-threshold step

$$\alpha_{t+1}^k = S_{\lambda/L}\left(\tilde{\alpha}_{t+1}^k\right).$$

where:

$$\mathcal{S}_\tau(v) = \mathrm{sign}(v)\max(|v| - \tau, 0).$$

after a predetermined number of iterations:

$$\alpha^{k+1} = \alpha_{t+1}^k.$$

It seems that we still need large D_{total} but in patch version we optimize the total α by optimizing individual:

$$\alpha_{\text{total}} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_L \end{bmatrix}$$

$$\alpha_\ell^{k+1} = \arg\min_\alpha \lambda\|\alpha\|_1 + \frac{\mu}{2}\left\|P_\ell z_x^{k+1} - D\alpha + u_\ell^k\right\|_2^2.$$

where

$$p_\ell^{k+1} = P_\ell z_x^{k+1}$$

Defining: $m_\ell^k := p_\ell^{k+1} + u_\ell^k$

and ISTA steps become:

$$\tilde{\alpha}_\ell^{(t+1)} = \alpha_\ell^{(t)} - \frac{\mu}{L} D^\top \left(D\alpha_\ell^{(t)} - m_\ell^k\right)$$

and

$$\alpha_\ell^{(t+1)} = S_{\frac{\lambda}{L}}\left(\tilde{\alpha}_\ell^{(t+1)}\right)$$

Optimizing u

We already know that (by the derived algorithm of ADMM, [8] [9])

$$u^{k+1} = u^k + \left(z_x^{k+1} - D_z\alpha^{k+1}\right)$$

Previously we have defined:

$$u_\ell^k := P_\ell u^k, \quad u_\ell^{k+1} := P_\ell u^{k+1}.$$

Knowing that:

$$D_z = \begin{bmatrix} P_1^\top D & P_2^\top D & \cdots & P_L^\top D \end{bmatrix} \in \mathbb{R}^{M\times LK}$$

$$D_z\alpha^{k+1} = \sum_{\ell=1}^{L} P_\ell^\top D\alpha_\ell^{k+1}$$

Multiplying the expression by P_ℓ

$$P_\ell u^{k+1} = P_\ell u^k + \left(P_\ell z_x^{k+1} - P_\ell D_z\alpha^{k+1}\right)$$

$$u_\ell^{k+1} = u_\ell^k + \left(P_\ell z_x^{k+1} - P_\ell D_z\alpha^{k+1}\right)$$

Remark:

$$P_\ell D_z \alpha^{k+1} = P_\ell W^{-1} \sum_{j=1}^{L} P_j^\top D \alpha_j^{k+1} = P_\ell W^{-1} \sum_{j=1}^{L} P_j^\top D \alpha_j^{k+1}$$

Then the generic solution is:

$$u_\ell^{k+1} = u_\ell^k + \left(P_\ell z_x^{k+1} - P_\ell z_{\text{dict}}^{k+1} \right)$$

where

$$z_{\text{dict}}^{k+1} = W^{-1} \sum_{j=1}^{L} P_j^\top D \alpha_j^{k+1}.$$

We can assume patches to be non-overlapping for ease of computation. Knowing that $P_\ell P_j^\top = \delta_{l,j} I_{n_p}$ for non-overlapping patches:

$$P_\ell D_z \alpha^{k+1} = D \alpha_\ell^{k+1}$$

If we do it for all P_ℓ , this global update is equivalent to so-called patch-wise update as

$$u_\ell^{k+1} = u_\ell^k + \left(P_\ell z_x^{k+1} - D \alpha_\ell^{k+1} \right)$$

3.4 Overall Algorithm Structure

The entire algorithm of our pipeline is presented in 3 major sections as seen below:

Algorithm 8 Kernel Estimation: Posterior Initialization + Cho–Lee Multi-Scale Refinement

Input: Training pairs $\{(x_m^{tr}, y_m^{tr})\}_{m=1}^{M}$; patch size p; kernel sizes $\{k_s\}$; ridge λ; prior scale β_{tr}; feature scale γ; KDE bandwidth constant A;
Test blurry image y; pyramid scales $\mathcal{S} = \{32, 64, 128, 256\}$; iterations per scale N_s;
Cho–Lee parameters $(\sigma_s, \sigma_r, \Delta t)$; weights (w_0, w_1, w_2); kernel reg. β; latent reg. α.
Output: Refined blur kernel $\hat{h}$ (and optionally latent $\hat{X}$ at full scale).

Stage 1A: build kernel library from training patches Extract overlapping patches $\{(x_{m,j}, y_{m,j})\}$ from $\{(x_m^{tr}, y_m^{tr})\}$ **for each** $(x_{m,j}, y_{m,j})$ **do**

Compute FFTs $X_{m,j}(\omega) = \mathcal{F}\{x_{m,j}\}$, $Y_{m,j}(\omega) = \mathcal{F}\{y_{m,j}\}$ Compute per-frequency kernel (ridge inversion): $H_{m,j}(\omega) \leftarrow \dfrac{Y_{m,j}(\omega)\, X_{m,j}^*(\omega)}{|X_{m,j}(\omega)|^2 + \lambda}$ Inverse FFT $\Rightarrow h_{m,j}$; center & crop to $k \times k$; project to PSF: $h_{m,j} \leftarrow \max(h_{m,j}, 0)$, $h_{m,j} \leftarrow h_{m,j} / \sum h_{m,j}$ Compute reconstruction loss $\ell_{m,j}$ and blur feature vector $\phi_{m,j} \in \mathbb{R}^4$ Store $\{h_i, \ell_i, \phi_i\}_{i=1}^{N}$

Stage 1B: prior weights and KDE bandwidth Shift losses $\tilde{\ell}_i = \ell_i - \min_j \ell_j$ Set $w_i^{prior} \leftarrow \dfrac{\exp(-\beta_{tr}\tilde{\ell}_i)}{\sum_j \exp(-\beta_{tr}\tilde{\ell}_j)}$ Set $n_h = k^2$ and $h_{opt} \leftarrow A\, N^{-1/(n_h+4)}$

Stage 2: posterior update + jittered sampling on test patches Extract non-overlapping test patches $\{y^{(k)}\}_{k=1}^{K}$ and features ϕ^k **for** $k = 1$ **to** K **do**

Compute distances $d_{i,k}^2 = \|\phi_i - \phi^k\|_2^2$ Posterior weights: $w_{i,k}^{post} \leftarrow \dfrac{w_i^{prior} \exp(-\gamma d_{i,k}^2)}{\sum_j w_j^{prior} \exp(-\gamma d_{j,k}^2)}$ Weighted mean and covariance: $\mu_k \leftarrow \sum_i w_{i,k}^{post} h_i$, $S_k \leftarrow \sum_i w_{i,k}^{post}(h_i - \mu_k)(h_i - \mu_k)^\top + \varepsilon I$ Sample ancestor $a_k \sim \mathrm{Cat}(w_{\cdot,k}^{post})$, draw $\epsilon_k \sim \mathcal{N}(0, I)$ Set $\tilde{h}_k \leftarrow h_{a_k} + h_{opt}\, D_k \epsilon_k$ where $S_k = D_k D_k^\top$ Reshape to $k \times k$, project to PSF constraints

Aggregate initial kernel: $\hat{h}^{(0)} \leftarrow \dfrac{1}{K}\sum_{k=1}^{K} \tilde{h}_k$; Project $\hat{h}^{(0)}$ to PSF constraints

Stage 3: Cho–Lee coarse-to-fine refinement Build pyramid $Y^{(s)} = \texttt{Resize}(y, s \times s)$ for $s \in \mathcal{S}$ Initialize at coarsest scale $s_0 = 32$: $X^{(s_0)} \leftarrow Y^{(s_0)}$, $h^{(s_0)} \leftarrow \texttt{Resize}(\hat{h}^{(0)}, k_{s_0} \times k_{s_0})$ **for each** $s \in \mathcal{S}$ **do**

if $s \neq s_0$ **then**

$X^{(s)} \leftarrow \texttt{Upsample}(X^{(prev)})$, $h^{(s)} \leftarrow \texttt{Resize}(h^{(prev)}, k_s \times k_s)$

for $t = 1$ **to** N_s **do**

Prediction (Cho–Lee): $X_b \leftarrow \texttt{Bilateral}(X^{(s)}; \sigma_s, \sigma_r)$ $X_s \leftarrow \texttt{Shock}(X_b; \Delta t)$ Compute gradients (g_x, g_y), magnitude $g = \sqrt{g_x^2 + g_y^2}$ Choose threshold τ so a fixed fraction survives; mask $\mathcal{M} = \{(i,j) : g(i,j) \geq \tau\}$ $P_x \leftarrow g_x \cdot \mathbf{1}_{\mathcal{M}}$, $P_y \leftarrow g_y \cdot \mathbf{1}_{\mathcal{M}}$

Kernel update (weighted LS in FFT): Form derivative channels $m \in \{x, y, xx, yy, xy\}$ with (P_m, Y_m) and weights w_m Update $\widehat{H}(\omega) \leftarrow \dfrac{\sum_m w_m \widehat{P}_m^*(\omega)\widehat{Y}_m(\omega)}{\sum_m w_m |\widehat{P}_m(\omega)|^2 + \beta}$ Inverse FFT $\Rightarrow h^{(s)}$; center/crop; threshold small values; project to PSF

Latent update (weighted LS in FFT): Compute $M_1(\omega) = |D_x|^2 + |D_y|^2$, $M_2(\omega) = |D_{xx}|^2 + |D_{yy}|^2 + |D_{xy}|^2$ $W(\omega) \leftarrow w_0 + w_1 M_1(\omega) + w_2 M_2(\omega)$ Update $\widehat{X}(\omega) \leftarrow \dfrac{\widehat{H}^*(\omega)\, \widehat{Y}^{(s)}(\omega)\, W(\omega)}{|\widehat{H}(\omega)|^2 W(\omega) + \alpha M_1(\omega) + \epsilon}$ Inverse FFT $\Rightarrow X^{(s)}$

$prev \leftarrow s$

return $\hat{h} \leftarrow h^{(256)}$ (and optionally $\hat{X} \leftarrow X^{(256)}$)

Algorithm 9 Dictionary Learning: Online Patch Dictionary (Mairal et al.)

Input: Sharp training images $\{x_m^{tr}\}_{m=1}^{M}$; patch operator $\{P_\ell\}$ with patch size $p \times p$; dictionary size K; sparsity weight λ; iterations T; initial dictionary $D_0 \in \mathbb{R}^{p^2 \times K}$.

Output: Learned patch dictionary $D \in \mathbb{R}^{p^2 \times K}$.

Initialize $D \leftarrow D_0$, $A \leftarrow 0 \in \mathbb{R}^{K \times K}$, $B \leftarrow 0 \in \mathbb{R}^{p^2 \times K}$

for $t = 1$ **to** T **do**

  Sample a sharp training patch $x_t \in \mathbb{R}^{p^2}$ from the pool of $\{P_\ell x_m^{tr}\}$

  Sparse coding (LASSO): $\alpha_t \leftarrow \arg\min_{\alpha \in \mathbb{R}^K} \frac{1}{2}\|x_t - D\alpha\|_2^2 + \lambda\|\alpha\|_1$

  Update sufficient statistics: $A \leftarrow A + \alpha_t \alpha_t^\top, \quad B \leftarrow B + x_t \alpha_t^\top$

  Dictionary update (column-wise, normalized): for $j = 1$ **to** K **do**

    $u_j \leftarrow \frac{1}{A[j,j]}\big(b_j - D a_j\big) + d_j \quad d_j \leftarrow \frac{u_j}{\max(\|u_j\|_2, 1)}$

return D

Algorithm 10 Inference: Deblur via Scaled ADMM with Fixed Kernel h and Learned Dictionary D

Input: Blurry image y (vectorized $z_y \in \mathbb{R}^M$); fixed kernel h with FFT $H(\omega)$; learned dictionary $D \in \mathbb{R}^{p^2 \times K}$; patch operators $\{P_\ell\}_{\ell=1}^L$;
ADMM penalty μ (optionally scheduled); sparsity weight λ; ISTA steps T_{ISTA} and step size L; max ADMM iters $K_{\max}$.
Output: Restored clean image $z_x^\star$ (and patch codes $\{\alpha_\ell\}$).

Initialize $z_x^0 \leftarrow z_y, \quad \alpha_\ell^0 \leftarrow 0, \quad u_\ell^0 \leftarrow 0$ for all ℓ
for $k = 0$ **to** $K_{\max} - 1$ **do**
  (1) z_x-update (FFT diagonalization of circular convolution) Form patch-fusion term (as in report): $b^k \leftarrow W^{-1} \sum_{\ell=1}^L P_\ell^\top \big(D\alpha_\ell^k - u_\ell^k\big)$ where $W = \sum_{\ell=1}^L P_\ell^\top P_\ell$ Compute FFTs: $Y(\omega) \leftarrow \mathcal{F}\{z_y\}$, $B^k(\omega) \leftarrow \mathcal{F}\{b^k\}$ Element-wise update for all frequencies ω:

$$X^{k+1}(\omega) \leftarrow \frac{H^*(\omega)Y(\omega) + \mu\, B^k(\omega)}{|H(\omega)|^2 + \mu}$$

  Inverse FFT: $z_x^{k+1} \leftarrow \mathcal{F}^{-1}\{X^{k+1}(\omega)\}$
  (2) α-update (patch-wise ISTA / soft-threshold) for each *patch* $\ell = 1, \ldots, L$ **do**
    $p_\ell^{k+1} \leftarrow P_\ell z_x^{k+1}, \quad m_\ell^k \leftarrow p_\ell^{k+1} + u_\ell^k \quad \alpha_\ell^{(0)} \leftarrow \alpha_\ell^k$ **for** $t = 0$ **to** $T_{ISTA} - 1$ **do**
      $\tilde{\alpha}_\ell^{(t+1)} \leftarrow \alpha_\ell^{(t)} - \frac{\mu}{L} D^\top \big(D\alpha_\ell^{(t)} - m_\ell^k\big) \quad \alpha_\ell^{(t+1)} \leftarrow \mathcal{S}_{\lambda/L}\big(\tilde{\alpha}_\ell^{(t+1)}\big)$
    $\alpha_\ell^{k+1} \leftarrow \alpha_\ell^{(T_{\text{ISTA}})}$
  (3) Dual update (scaled form) for each *patch* $\ell = 1, \ldots, L$ **do**
    $u_\ell^{k+1} \leftarrow u_\ell^k + \big(P_\ell z_x^{k+1} - D\alpha_\ell^{k+1}\big)$
  Stopping (optional): stop if primal/dual residuals satisfy tolerances
return $z_x^\star \leftarrow z_x^{k+1}$

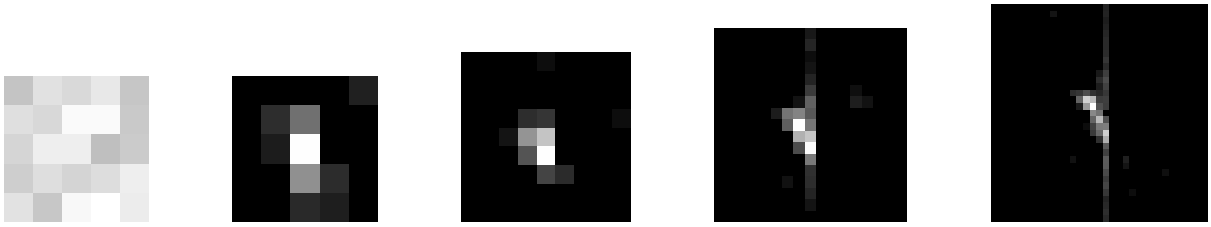
4 Results

4.1 Kernel Estimation

End-to-end pipeline for the experiment can be described as follows:

Based on the samples from the kernels we build the prior and we apply the posterior weight update corresponding to posterior distribution. The samples from it will initialize the iterative algorithm of Cho-Lee, which runs for 12 iterations.

The results are depicted in the Fig 1, which shows that the end result captures the motion.



(a) Initial estimation sample (b) 5x5 estimation (c) 7x7 estimation (d) 9x9 estimation (e) 11x11 estimation

Figure 1: Evolution of the estimated kernels throughout Cho and Lee of the image shown in Fig 3.

4.2 Dictionary Reconstruction and the Results of the ADMM

We use the predicted H and fix it throughout ADMM as a design choice. For the iterative steps of ADMM, initially we start with large dictionary constraint hyperparameter μ so that the estimated clear image update depends heavily on the dictionary reconstruction $\|z_x - D_z\alpha\|_2^2$, which promotes the iteration outcomes to lie in the clear image space (i.e which overall can be interpreted as avoidance of the trivial blurred prediction for the clear image, encouraged by the separate h and x optimization and dictionary reconstruction). The outcomes of the ADMM steps indicate the the predicted image gets cleared as steps increased as can be seen in Fig 3 and 4. As can be seen on the Fig 2, dictionary can reconstruct the clear image.



(a) Input blurred image (b) Dictionary-based reconstruction (c) Final deblurred result

Figure 2: Deblurring using dictionary-based reconstruction on the building image.

We claim that the sharpness of the image can be measured by high frequency components and graident based metrics (such as Tenengrad $= \sum_{x,y}\left(G_x^2 + G_y^2\right)$) and the metrics are shown in the Tables 1 and 2



(a) Blurred Input



(b) ADMM iteration 3



(c) ADMM iteration 10



(d) Final ADMM result

Figure 3: Evolution of the ADMM reconstruction across selected iterations and the final result.

Moreover, here is the radial FFT power plots to explicitly see how high frequencies are reintroduced after deblurring as can be seen in the Fig 5.



(a) Input blurred image



(b) ADMM iteration 3

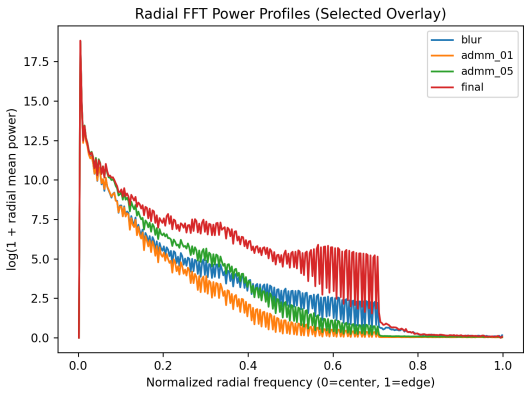


(c) ADMM iteration 10

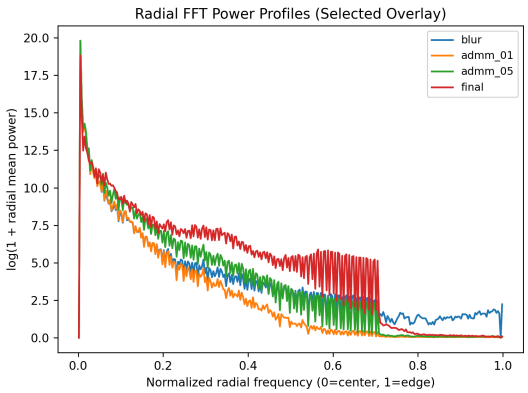


(d) Final ADMM reconstruction

Figure 4: ADMM deblurring process showing the input blurred image, intermediate reconstructions, and the final result.



(a) Cup



(b) Building

Figure 5: Radial FFT comparison between the Cup and Building Images.

Table 1: Sharpness-related metrics across ADMM iterations and final output (of Fig 4).

idx	label	file	grad_power	tenengrad	hf_ratio
0	blur	input_blur.png	5.944804e-04	5.944804e-04	0.000861
1	admm_00	admm_00.png	2.491733e-04	2.491733e-04	0.000255
2	admm_01	admm_01.png	5.062831e-04	5.062831e-04	0.000146
3	admm_02	admm_02.png	7.974522e-04	7.974522e-04	0.000214
4	admm_03	admm_03.png	1.051128e-03	1.051128e-03	0.000313
5	admm_04	admm_04.png	1.275241e-03	1.275241e-03	0.000442
6	admm_05	admm_05.png	1.485745e-03	1.485745e-03	0.000598
7	admm_06	admm_06.png	1.690356e-03	1.690356e-03	0.000786
8	admm_07	admm_07.png	1.895307e-03	1.895307e-03	0.001009
9	admm_08	admm_08.png	2.105915e-03	2.105915e-03	0.001276
10	admm_09	admm_09.png	2.326757e-03	2.326757e-03	0.001605
11	admm_10	admm_10.png	2.560284e-03	2.560284e-03	0.002021
12	admm_11	admm_11.png	2.805360e-03	2.805360e-03	0.002558
13	admm_12	admm_12.png	3.061679e-03	3.061679e-03	0.003263
14	admm_13	admm_13.png	3.325560e-03	3.325560e-03	0.004183
15	admm_14	admm_14.png	3.599105e-03	3.599105e-03	0.005386
16	admm_15	admm_15.png	3.871754e-03	3.871754e-03	0.006822
17	admm_16	admm_17.png	4.317180e-03	4.317180e-03	0.009264
18	admm_17	admm_18.png	4.530648e-03	4.530648e-03	0.010648
19	admm_18	admm_19.png	4.742994e-03	4.742994e-03	0.012195
20	final	final_result.png	4.742994e-03	4.742994e-03	0.012195

Table 2: Sharpness metrics computed over ADMM iterations and the final reconstructio (of Fig 3) .

idx	label	file	grad_power	tenengrad	hf_ratio
0	blur	input_blur.png	5.694281e-04	5.694281e-04	0.000473
1	admm_00	admm_00.png	3.109234e-04	3.109234e-04	0.000203
2	admm_01	admm_01.png	6.035091e-04	6.035091e-04	0.000092
3	admm_02	admm_02.png	9.359412e-04	9.359412e-04	0.000138
4	admm_03	admm_03.png	1.230954e-03	1.230954e-03	0.000242
5	admm_04	admm_04.png	1.482488e-03	1.482488e-03	0.000396
6	admm_05	admm_05.png	1.683529e-03	1.683529e-03	0.000573
7	admm_06	admm_06.png	1.862015e-03	1.862015e-03	0.000800
8	admm_07	admm_07.png	2.027196e-03	2.027196e-03	0.001072
9	admm_08	admm_08.png	2.186032e-03	2.186032e-03	0.001383
10	admm_09	admm_09.png	2.339068e-03	2.339068e-03	0.001728
11	admm_10	admm_10.png	2.481365e-03	2.481365e-03	0.002090
12	admm_11	admm_11.png	2.609914e-03	2.609914e-03	0.002448
13	admm_12	admm_12.png	2.723556e-03	2.723556e-03	0.002796
14	admm_13	admm_13.png	2.822662e-03	2.822662e-03	0.003120
15	admm_14	admm_14.png	2.908497e-03	2.908497e-03	0.003420
16	admm_15	admm_15.png	2.981219e-03	2.981219e-03	0.003691
17	admm_16	admm_16.png	3.044691e-03	3.044691e-03	0.003947
18	admm_17	admm_17.png	3.103093e-03	3.103093e-03	0.004201
19	admm_18	admm_18.png	3.157139e-03	3.157139e-03	0.004460
20	admm_19	admm_19.png	3.209012e-03	3.209012e-03	0.004732
21	final	final_result.png	4.742994e-03	4.742994e-03	0.012195

4.3 Wiener vs ADMM

We then compared visually the outputs of the baseline Wiener filter (using our estimated kernel to obtain the blur image.

As can be seen in the Fig 6 , the Wiener filter performs worse when it comes to especially



(a) Input blurred image (b) Wiener deblurring result (c) ADMM deblurring result

Figure 6: Comparison of input blur, Wiener deblurring, and ADMM-based deblurring.

5 Discussion

Our pipeline begins with kernel estimation, which is one of the most crucial part of the blind image de-blurring, since throughout our experiments, we have observed that the correct estimation of kernel affects the rest of the algorithms dramatically. In our method, we initialize the kernel in 5*5 dimensions, which appears to be engulfed by the motion blur (many white points), which can be avoided with larger size initialization. Nonetheless, Cho and Lee based approach fine tunes the kernel and clearly shows the alignment with the motion gradually.

Based on our sharpness evaluation metrics, we clearly see that the ADMM steps gradually increase sharpness as verified by the visual outputs too. Our learned patch-based dictionary shows its reconstruction capability. Moreover, experimented Wiener filter can under-perform our algorithm.

However, our algorithm ADMM is based on FFT domain to facilitate the updates, which implicitly assumes circular convolution model, which gives us wrap-around stripes at the edges. Additionally, adjusting the sweet-spot of the hyper-parameters is a lengthy process since our algorithm is slower due to many sub-steps such as dictionary learning, ISTA iterations inside the ADMM. Moreover, the performance depends highly on the dataset, from which we obtain the initialization of the kernel estimation and we learn the dictionary.

6 Conclusion

In this project, we aimed to address the deblurring of an image, by first estimating the blur kernel via posterior sample initialization and multi-scale fine tuning, online patch dictionary learning for the sparse representation, indicating the clear image subspace and patch-wise ADMM iterative algorithm to gradually obtain the clear image. Hence, the method remains scalable to large datasets and high-resolution images. However, FFT wraps-around should be suppressed and the further fine tuning of various hyperparameters to improve the performance of dictionary learning, ADMM and inside ISTA steps should be performed. This requires a lengthy careful fine tuning due to having many hyperparameter and longer ADMM iteration duration (when compared to classical Wiener). Initial kernel estimation and dictionary learning heavily rely on the utilized dataset. Thus, any non-ideal and incompatible dataset can give misleading results.

Although the entire algorithm requires to be further fine-tuned, the idea of learning a clear image subspace via Dictionary using as much dataset as we want and eventually turning our de-blurring task to an

optimization problem with patch-wise feasible implementation builds the foundation of our project. Additionally, optimization based solution is about image restoration problem, agnostic to specific domains, this gives a solution for the cases outside the common de-blurring, as long as the dataset is available.

7 Appendix

Code Repository

The codes and the full resolution file of this report are available at the following Google Drive folder:

https://drive.google.com/drive/folders/184Hr_ovZ_9c13t4jfxK1Y6azXnRJB092?usp=sharing

Files included:

- `Original Report File`
- `evaluate-metrics.py -> Performance metric evaluation`
- `Motion_deblurring_main_code.py -> Main algorithm`
- `wiener-test.py -> Wiener filtering using the estimated kernel for comparison purposes`

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